

SALES DATA ANALYSIS REPORT

Dataset: retail_sales_test_dataset_6500plus.csv | Analysis Period: 2021-01-01 → 2025-12-31 | Generated: August 16, 2026 11:52 AM

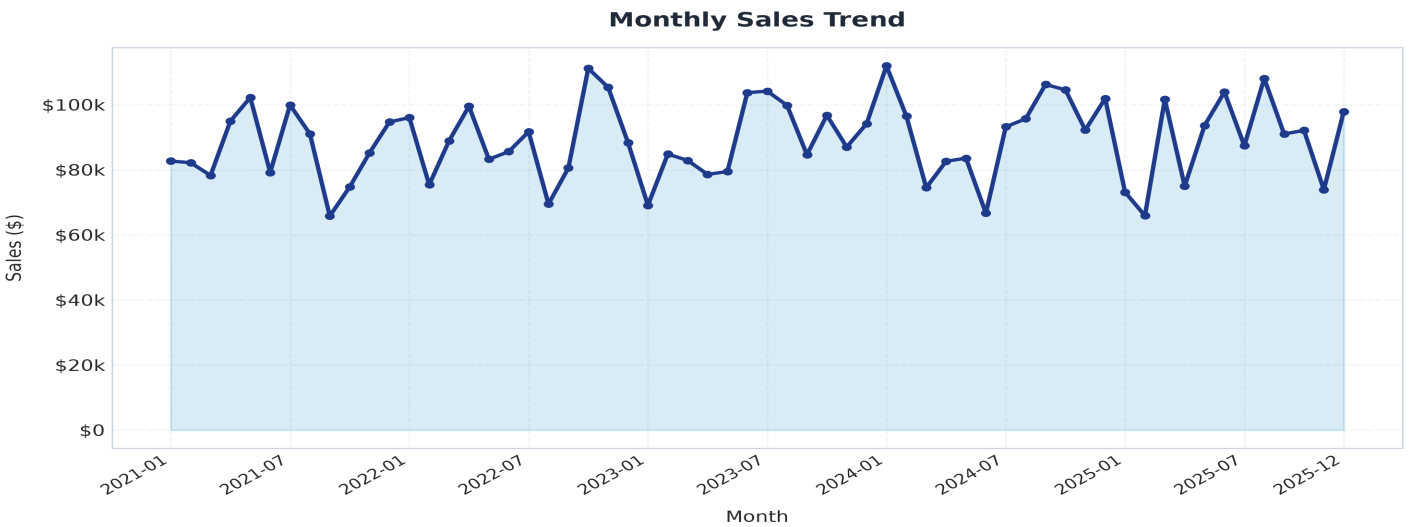
1. Executive Summary & Core KPIs

TOTAL REVENUE	TOTAL UNITS SOLD	TRANSACTIONS	AVG ORDER VALUE
\$5,345,862.68	15,538	6,488	\$823.96
TOP PRODUCT	TOP CATEGORY	TOP REGION	SALES TREND
Laptop Pro 15	Technology	North	Increasing (+5.2%)

2. Dataset Overview & Data Quality Audit

Quality Metric	Raw Ingestion	Post-Cleaning	Operational Resolution
Total Records	6,560	6,488	Invalid dates and unrecoverable records pruned
Duplicate Rows	60	0	Exact record duplicates identified and removed
Missing Values	200	0	Numeric imputed with median; Categories with mode
Invalid Dates	12	0	Unparseable dates dropped to preserve time series

3. Monthly Sales Trajectory

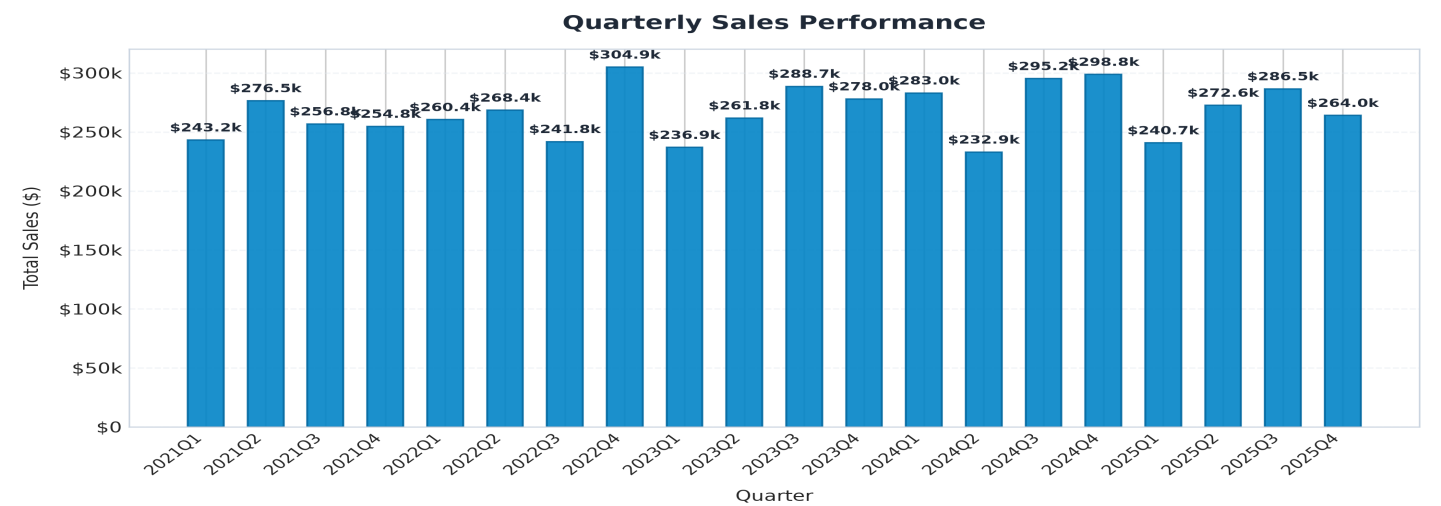


Recent Monthly Breakdown (Latest Periods):

Month	Sales (\$)	Quantity	Avg Order (\$)	Transactions
2025-07	\$87,410.16	272	\$773.54	113
2025-08	\$108,100.51	333	\$745.52	145
2025-09	\$91,038.33	244	\$901.37	101
2025-10	\$92,146.50	223	\$949.96	97
2025-11	\$73,916.62	239	\$794.80	93

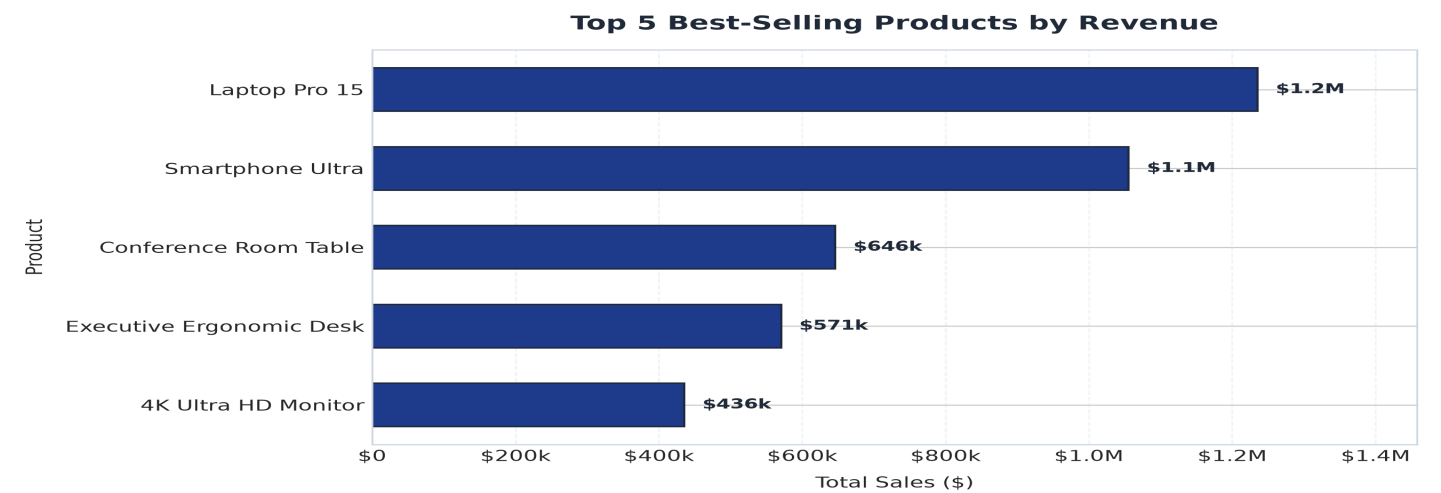
2025-12	\$97,909.55	302	\$874.19	112
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4. Quarterly & Product Performance

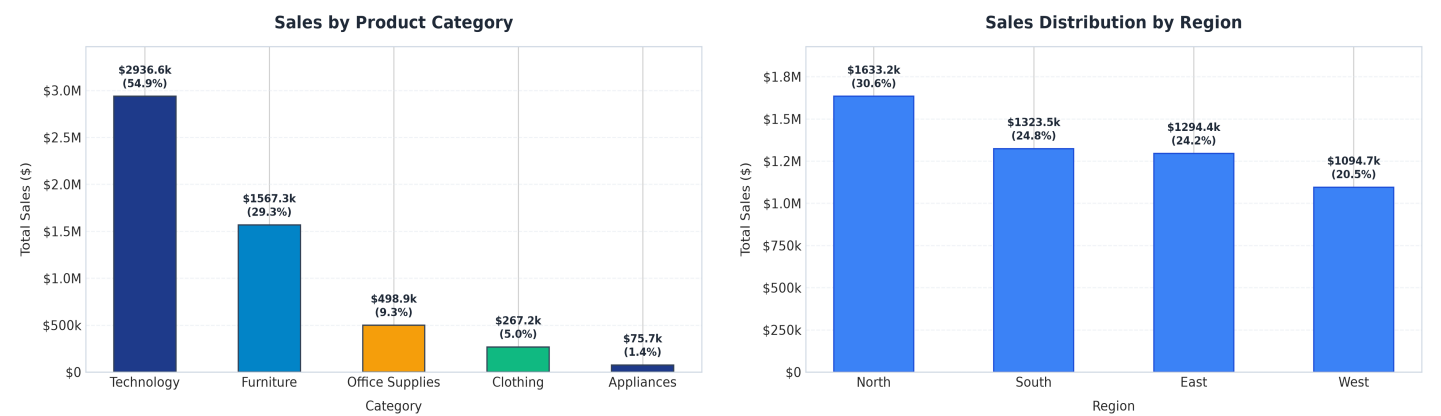


Top Best-Selling Products by Revenue:

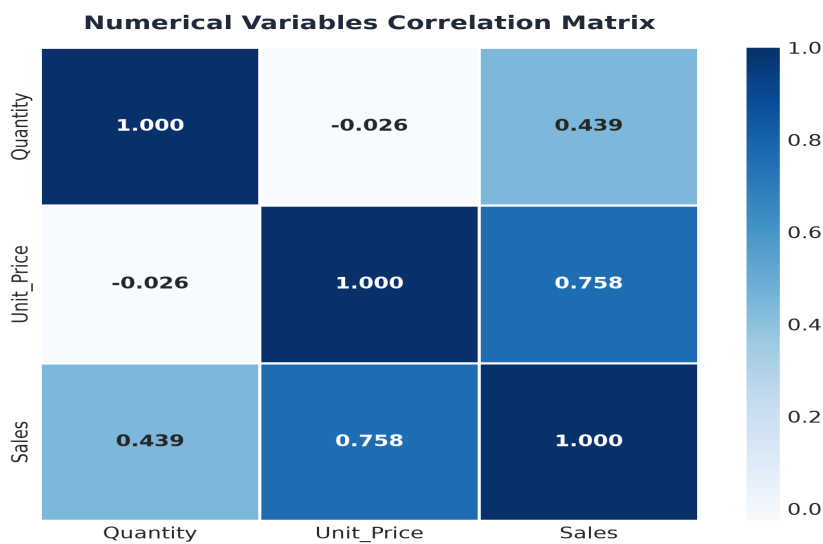
Rank	Product Name	Category	Total Revenue	Units Sold	Avg Unit Price
#1	Laptop Pro 15	Technology	\$1,235,554.33	1,033	\$1,198.67
#2	Smartphone Ultra	Technology	\$1,055,646.79	1,245	\$849.02
#3	Conference Room Table	Furniture	\$646,146.82	805	\$802.54
#4	Executive Ergonomic Desk	Furniture	\$570,671.23	1,023	\$557.93
#5	4K Ultra HD Monitor	Technology	\$435,606.20	1,208	\$360.85



5. Category & Regional Distribution



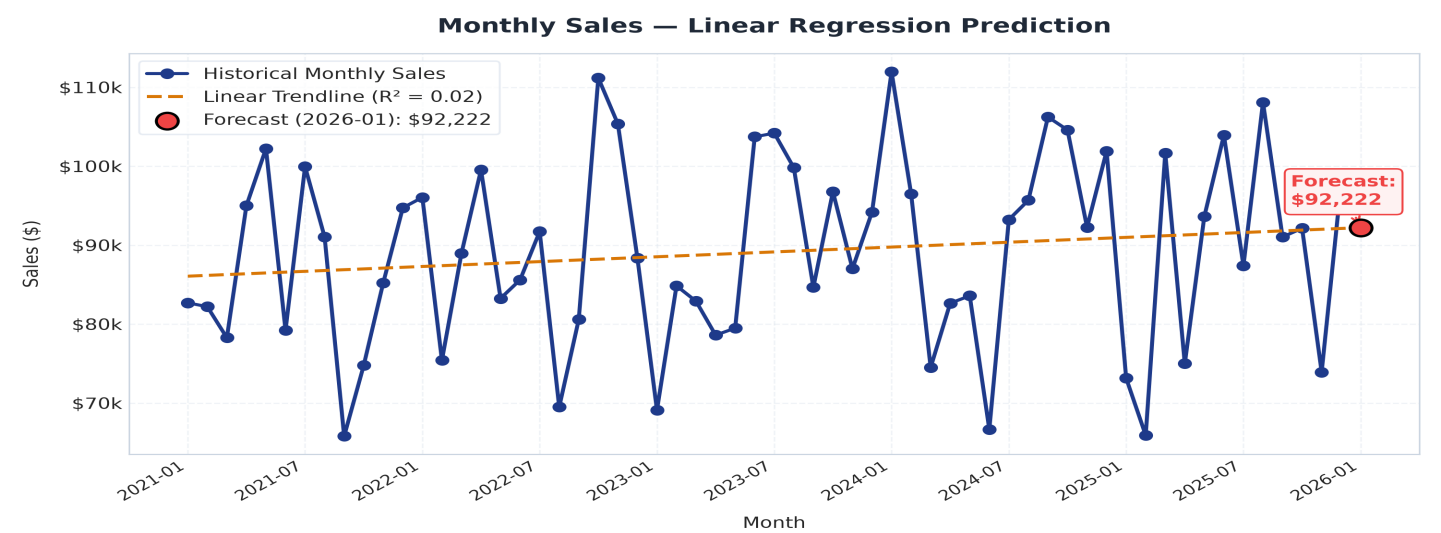
6. Numerical Variable Correlation Analysis



Correlation Insights: In the analyzed dataset, the Pearson correlation between Unit Price and Sales is **0.758**, while Quantity sold and Sales exhibits a correlation of **0.439**. Unit Price represents a stronger revenue driver than order volume alone.

7. Predictive Sales Forecasting (Simple Linear Regression)

Target Forecast Period	Predicted Revenue (\$)	Model Fit (R² Score)	Monthly Trajectory Slope
2026-01	\$92,222.46	0.0221	\$+102.45 / mo



Notice & Model Disclaimer: This prediction is based on a simple linear regression model and should be treated as a basic trend estimate rather than a production-grade forecasting model.

8. Data-Driven Business Insights & Conclusion

- 1. Financial Performance:** Over the evaluated period (2021-01-01 to 2025-12-31), the business logged **\$5,345,862.68** across **6,488** orders with an average sale of **\$823.96**.
- 2. Catalog Drivers:** Product **Laptop Pro 15** generated top revenue within the **Technology** merchandise category.
- 3. Territory Performance:** Region **North** represents the top sales territory.
- 4. Forecast Trajectory:** Sequential OLS regression forecasts revenue of **\$92,222.46** for **2026-01** with a monthly rate of change of **\$+102.45/mo**.